## 1. Store Student Details in a Map

```
void main() {
  Map<String, dynamic> student = {
    'Name': 'Gulshan',
    'Roll No': 101,
    'Course': 'BCA'
  };

  student.forEach((key, value) {
    print('$key: $value');
  });
}
```

## 2. Convert Student Object to Map and Back

```
class Student {
  String name;
  int rollNo;

  Student(this.name, this.rollNo);

  Map<String, dynamic> toMap() {
    return {'name': name, 'rollNo': rollNo};
  }

  factory Student.fromMap(Map<String, dynamic> map) {
    return Student(map['name'], map['rollNo']);
  }
}

void main() {
  Student s1 = Student('Gulshan', 101);
  Map<String, dynamic> map = s1.toMap();
  Student s2 = Student.fromMap(map);

  print(map);
  print('${s2.name} ${s2.rollNo}');
}
```

## 3. Convert Map to JSON and JSON to Map

```
import 'dart:convert';

void main() {
  Map<String, dynamic> student = {
    'name': 'Gulshan',
    'rollNo': 101
  };

  String jsonData = jsonEncode(student);
  print(jsonData);

  Map<String, dynamic> data = jsonDecode(jsonData);
  print(data);
}
```

## 4. Save User Name Using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Home()));

class Home extends StatefulWidget {
  @override
  State<Home> createState() => _HomeState();
}

class _HomeState extends State<Home> {
  final name = TextEditingController();
  String savedName = '';

  @override
  void initState() {
    super.initState();
    loadName();
  }

  Future<void> loadName() async {
    final p = await SharedPreferences.getInstance();
    setState(() => savedName = p.getString('name') ?? '');
  }

  Future<void> saveName() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('name', name.text);
    await loadName();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Save Name')),
      body: Column(children: [
        TextField(controller: name),
        ElevatedButton(
          onPressed: saveName,
          child: Text('Save'),
        ),
        Text('Welcome $savedName'),
      ]),
    );
  }
}
```

## 5. Store and Retrieve Counter Using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: CounterApp()));

class CounterApp extends StatefulWidget {
  @override
  State<CounterApp> createState() => _CounterAppState();
}

class _CounterAppState extends State<CounterApp> {
  int count = 0;

  @override
  void initState() {
    super.initState();
    loadCount();
  }

  Future<void> loadCount() async {
    final p = await SharedPreferences.getInstance();
    setState(() => count = p.getInt('count') ?? 0);
  }

  Future<void> increment() async {
    final p = await SharedPreferences.getInstance();
    setState(() => count++);
    await p.setInt('count', count);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Counter')),
      body: Center(
        child: Column(mainAxisSize: MainAxisSize.min, children: [
          Text('Count: $count'),
          ElevatedButton(
            onPressed: increment,
            child: Text('Increment'),
          ),
        ]),
      ),
    );
  }
}
```

## 6. Save Dark Mode ON/OFF Using a Switch

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: DarkApp()));

class DarkApp extends StatefulWidget {
  @override
  State<DarkApp> createState() => _DarkAppState();
}

class _DarkAppState extends State<DarkApp> {
  bool dark = false;

  @override
  void initState() {
    super.initState();
    loadMode();
  }

  Future<void> loadMode() async {
    final p = await SharedPreferences.getInstance();
    setState(() => dark = p.getBool('dark') ?? false);
  }

  Future<void> changeMode(bool value) async {
    final p = await SharedPreferences.getInstance();
    await p.setBool('dark', value);
    setState(() => dark = value);
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: dark ? ThemeData.dark() : ThemeData.light(),
      home: Scaffold(
        appBar: AppBar(title: Text('Dark Mode')),
        body: SwitchListTile(
          title: Text('Dark Mode'),
          value: dark,
          onChanged: changeMode,
        ),
      ),
    );
  }
}
```

## 7. Remember Me Checkbox Using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: LoginApp()));

class LoginApp extends StatefulWidget {
  @override
  State<LoginApp> createState() => _LoginAppState();
}

class _LoginAppState extends State<LoginApp> {
  bool remember = false;

  @override
  void initState() {
    super.initState();
    loadRemember();
  }

  Future<void> loadRemember() async {
    final p = await SharedPreferences.getInstance();
    setState(() => remember = p.getBool('remember') ?? false);
  }

  Future<void> saveRemember(bool value) async {
    final p = await SharedPreferences.getInstance();
    await p.setBool('remember', value);
    setState(() => remember = value);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Login')),
      body: Column(children: [
        TextField(decoration: InputDecoration(labelText: 'Username')),
        TextField(
          obscureText: true,
          decoration: InputDecoration(labelText: 'Password'),
        ),
        CheckboxListTile(
          title: Text('Remember Me'),
          value: remember,
          onChanged: (v) => saveRemember(v!),
        ),
        ElevatedButton(onPressed: () {}, child: Text('Login')),
      ]),
    );
  }
}
```

## 8. Save Name, Email and Age Using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: UserApp()));

class UserApp extends StatefulWidget {
  @override
  State<UserApp> createState() => _UserAppState();
}

class _UserAppState extends State<UserApp> {
  String details = '';

  @override
  void initState() {
    super.initState();
    loadData();
  }

  Future<void> saveData() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('name', 'Gulshan');
    await p.setString('email', 'gulshan@example.com');
    await p.setInt('age', 20);
    await loadData();
  }

  Future<void> loadData() async {
    final p = await SharedPreferences.getInstance();
    setState(() {
      details = '${p.getString('name') ?? ''}\n'
          '${p.getString('email') ?? ''}\n'
          '${p.getInt('age') ?? 0}';
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('User Details')),
      body: Column(children: [
        ElevatedButton(onPressed: saveData, child: Text('Save Details')),
        Text(details),
      ]),
    );
  }
}
```

## 9. Save a List of Favourite Subjects

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: SubjectsApp()));

class SubjectsApp extends StatefulWidget {
  @override
  State<SubjectsApp> createState() => _SubjectsAppState();
}

class _SubjectsAppState extends State<SubjectsApp> {
  List<String> subjects = [];

  @override
  void initState() {
    super.initState();
    loadSubjects();
  }

  Future<void> saveSubjects() async {
    final p = await SharedPreferences.getInstance();
    await p.setStringList('subjects', ['Python', 'Flutter', 'Java']);
    await loadSubjects();
  }

  Future<void> loadSubjects() async {
    final p = await SharedPreferences.getInstance();
    setState(() => subjects = p.getStringList('subjects') ?? []);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Favourite Subjects')),
      body: Column(children: [
        ElevatedButton(onPressed: saveSubjects, child: Text('Save Subjects')),
        ...subjects.map((s) => ListTile(title: Text(s))),
      ]),
    );
  }
}
```

## 10. Remove SharedPreferences Data Using a Clear Button

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: ClearApp()));

class ClearApp extends StatefulWidget {
  @override
  State<ClearApp> createState() => _ClearAppState();
}

class _ClearAppState extends State<ClearApp> {
  String name = '';

  @override
  void initState() {
    super.initState();
    loadData();
  }

  Future<void> saveData() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('name', 'Gulshan');
    await loadData();
  }

  Future<void> loadData() async {
    final p = await SharedPreferences.getInstance();
    setState(() => name = p.getString('name') ?? 'No Data');
  }

  Future<void> clearData() async {
    final p = await SharedPreferences.getInstance();
    await p.clear();
    await loadData();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Clear Data')),
      body: Column(children: [
        Text(name),
        ElevatedButton(onPressed: saveData, child: Text('Save')),
        ElevatedButton(onPressed: clearData, child: Text('Clear')),
      ]),
    );
  }
}
```

## 11. Show Welcome Screen Only on First Launch

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: WelcomeApp()));

class WelcomeApp extends StatefulWidget {
  @override
  State<WelcomeApp> createState() => _WelcomeAppState();
}

class _WelcomeAppState extends State<WelcomeApp> {
  String message = 'Loading...';

  @override
  void initState() {
    super.initState();
    checkLaunch();
  }

  Future<void> checkLaunch() async {
    final p = await SharedPreferences.getInstance();
    bool first = p.getBool('first') ?? true;

    setState(() {
      message = first ? 'Welcome to the App!' : 'Welcome Back!';
    });

    await p.setBool('first', false);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Welcome')),
      body: Center(child: Text(message)),
    );
  }
}
```

## 12. Write Text into a File

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

void main() => runApp(MaterialApp(home: FileApp()));

class FileApp extends StatelessWidget {
  Future<void> writeFile() async {
    final dir = await getApplicationDocumentsDirectory();
    final file = File('${dir.path}/data.txt');
    await file.writeAsString('Hello Flutter!');
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Write File')),
      body: Center(
        child: ElevatedButton(
          onPressed: writeFile,
          child: Text('Write Text'),
        ),
      ),
    );
  }
}
```

## 13. Read Text from a Stored File

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

void main() => runApp(MaterialApp(home: ReadApp()));

class ReadApp extends StatefulWidget {
  @override
  State<ReadApp> createState() => _ReadAppState();
}

class _ReadAppState extends State<ReadApp> {
  String text = '';

  Future<void> readFile() async {
    final dir = await getApplicationDocumentsDirectory();
    final file = File('${dir.path}/data.txt');

    if (await file.exists()) {
      text = await file.readAsString();
    } else {
      text = 'File not found';
    }

    setState(() {});
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Read File')),
      body: Column(children: [
        ElevatedButton(onPressed: readFile, child: Text('Read Text')),
        Text(text),
      ]),
    );
  }
}
```

## 14. Save a Student Object as JSON in SharedPreferences

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: StudentApp()));

class StudentApp extends StatefulWidget {
  @override
  State<StudentApp> createState() => _StudentAppState();
}

class _StudentAppState extends State<StudentApp> {
  String details = '';

  @override
  void initState() {
    super.initState();
    loadStudent();
  }

  Future<void> saveStudent() async {
    final p = await SharedPreferences.getInstance();
    String json = jsonEncode({
      'name': 'Gulshan',
      'roll': 101,
      'course': 'BCA'
    });
    await p.setString('student', json);
    await loadStudent();
  }

  Future<void> loadStudent() async {
    final p = await SharedPreferences.getInstance();
    String? json = p.getString('student');

    if (json != null) {
      final student = jsonDecode(json);
      setState(() {
        details = '${student['name']}\n'
            '${student['roll']}\n'
            '${student['course']}';
      });
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Student JSON')),
      body: Column(children: [
        ElevatedButton(onPressed: saveStudent, child: Text('Save Student')),
        Text(details),
      ]),
    );
  }
}
```

## 15. Simple To-Do Application Using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: TodoApp()));

class TodoApp extends StatefulWidget {
  @override
  State<TodoApp> createState() => _TodoAppState();
}

class _TodoAppState extends State<TodoApp> {
  final controller = TextEditingController();
  List<String> tasks = [];

  @override
  void initState() {
    super.initState();
    loadTasks();
  }

  Future<void> loadTasks() async {
    final p = await SharedPreferences.getInstance();
    setState(() => tasks = p.getStringList('tasks') ?? []);
  }

  Future<void> addTask() async {
    if (controller.text.trim().isEmpty) return;

    tasks.add(controller.text.trim());
    final p = await SharedPreferences.getInstance();
    await p.setStringList('tasks', tasks);

    controller.clear();
    await loadTasks();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('To-Do App')),
      body: Column(children: [
        TextField(controller: controller),
        ElevatedButton(onPressed: addTask, child: Text('Add Task')),
        Expanded(
          child: ListView(
            children: tasks.map((t) => ListTile(title: Text(t))).toList(),
          ),
        ),
      ]),
    );
  }
}
```

## 16. Notes Application Using a JSON File

```
import 'dart:convert';
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

void main() => runApp(MaterialApp(home: NotesApp()));

class NotesApp extends StatefulWidget {
  @override
  State<NotesApp> createState() => _NotesAppState();
}

class _NotesAppState extends State<NotesApp> {
  final controller = TextEditingController();
  List<String> notes = [];

  Future<File> getFile() async {
    final dir = await getApplicationDocumentsDirectory();
    return File('${dir.path}/notes.json');
  }

  Future<void> loadNotes() async {
    final file = await getFile();

    if (await file.exists()) {
      final data = jsonDecode(await file.readAsString());
      setState(() => notes = List<String>.from(data));
    }
  }

  Future<void> saveNote() async {
    if (controller.text.trim().isEmpty) return;

    notes.add(controller.text.trim());
    final file = await getFile();
    await file.writeAsString(jsonEncode(notes));

    controller.clear();
    setState(() {});
  }

  @override
  void initState() {
    super.initState();
    loadNotes();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Notes App')),
      body: Column(children: [
        TextField(controller: controller),
        ElevatedButton(onPressed: saveNote, child: Text('Save Note')),
        Expanded(
          child: ListView(
            children: notes.map((n) => ListTile(title: Text(n))).toList(),
          ),
        ),
      ]),
    );
  }
}
```

## 17. Student Profile Application

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: ProfileApp()));

class ProfileApp extends StatefulWidget {
  @override
  State<ProfileApp> createState() => _ProfileAppState();
}

class _ProfileAppState extends State<ProfileApp> {
  final name = TextEditingController();
  final roll = TextEditingController();
  final course = TextEditingController();
  final semester = TextEditingController();

  String profile = '';

  @override
  void initState() {
    super.initState();
    loadProfile();
  }

  Future<void> saveProfile() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('name', name.text);
    await p.setString('roll', roll.text);
    await p.setString('course', course.text);
    await p.setString('semester', semester.text);
    await loadProfile();
  }

  Future<void> loadProfile() async {
    final p = await SharedPreferences.getInstance();

    setState(() {
      name.text = p.getString('name') ?? '';
      roll.text = p.getString('roll') ?? '';
      course.text = p.getString('course') ?? '';
      semester.text = p.getString('semester') ?? '';

      profile = name.text.isEmpty
          ? 'No Profile Saved'
          : 'Name: ${name.text}\n'
            'Roll: ${roll.text}\n'
            'Course: ${course.text}\n'
            'Semester: ${semester.text}';
    });
  }

  Future<void> clearProfile() async {
    final p = await SharedPreferences.getInstance();
    await p.remove('name');
    await p.remove('roll');
    await p.remove('course');
    await p.remove('semester');

    name.clear();
    roll.clear();
    course.clear();
    semester.clear();
    await loadProfile();
  }

  Widget input(String label, TextEditingController c) {
    return TextField(
      controller: c,
      decoration: InputDecoration(labelText: label),
    );
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Student Profile')),
      body: SingleChildScrollView(
        padding: EdgeInsets.all(16),
        child: Column(children: [
          input('Name', name),
          input('Roll Number', roll),
          input('Course', course),
          input('Semester', semester),
          ElevatedButton(
            onPressed: saveProfile,
            child: Text('Save / Update'),
          ),
          ElevatedButton(
```

```
            onPressed: clearProfile,
            child: Text('Clear Profile'),
          ),
          SizedBox(height: 20),
          Text(profile),
        ]),
      ),
    );
  }
}
```